Five-anomaly model evaluation

Complete evaluation and improvement history • 13–17 October 2026

Recognition improved in selected domains. Reliable camera transfer is still unresolved.

6

Visual fine-tunes

2 CLIP encoders × 3 seeds

330 + 50

Strategy / video routes

New heads and routes; different studies

38 + tiles

Latest improvements

Dev-only policies + causal state replay

Strongest next investment: verified event histories and matched normal footage on independent target cameras.

Five objectives require different evidence

Existing dataset annotations and definitions remain the source of truth

Objective	Evidence required	Current qualification
Smoking	Person + hand/object/action evidence	Low-resolution CCTV differs from photos
Littering	Disposal action and scene context	IWDD includes larger waste dumping
Loitering	Identified person and timed history	A visible person alone is insufficient
Rough sleeping	Persistent apparent public sleep	Posture photos are provisional proxies
Unattended items	Item persistence + owner separation	30 s criterion belongs to AVSS source

Unknown history or missing labels stay unknown; other objectives are never automatically labelled normal.

Four image models and a separate VadCLIP study

All five were investigated; later adaptation focuses on the two CLIP encoders

Model	Evaluation / adaptation	Scope of conclusions
MobileCLIP2-S2	Zero-shot → probe → visual fine-tune	All five; region / temporal pilots
SigLIP2-B/16-224	Zero-shot → probe → visual fine-tune	All five; region / temporal pilots
SmolVLM-500M-Instruct	Deterministic prompted responses	Expanded four-model image study
MiniCPM-V 4.0	Unquantized prompted responses	Expanded four-model image study
VadCLIP	Separate supervised smoking adapter	Offline protocol; different supervision

VLMs were not fine-tuned in later rounds. Head fitting and visual-backbone updates are reported separately.

The work progressed from prompts to event evidence

These counts describe separate experiments, often reusing the same source footage

Stage	Completed work	Evidence produced
13 Oct	4 models; 13,400 predictions	Initial audit and VadCLIP branch
14 Oct	55,156 expanded predictions	Larger cohorts; qualified weak diagnostics
15–16 Oct	6 actual CLIP visual fine-tunes	Changed weights; three seeds each
16 Oct	330 configurations; 50 pilot heads	Data transfer, crops, temporal / tracks
17 Oct	38 policies; state replay; 830 views	Ensembles, memory, tiled item detections

New runs and artifacts preserve the earlier reports and slide decks.

Many files did not provide complete event labels

The dataset audit changed what could be evaluated honestly

Usable supervised evidence

Smoking: CCTV person/action labels
IWDD: dumping action clips + controls
Street Scene: person track/type labels
CAVIAR: item/person source boxes
ABODA: reviewed time intervals

Useful diagnostics, weaker event truth

6,667 littering candidate images
1,675 person/loitering candidates
Synthetic lying / falling postures
Static public-sleep photographs
MEVA abandonment-action windows

Positive-only images, augmented copies and pose labels cannot establish five-event precision or false-alarm rates.

Training covered all five targets with uneven evidence

First supervised study: 7,498 examples • 12,478 distinct view images

Objective	Train (normal / +)	Dev (normal / +)	Test (normal / +)
Smoking, CCTV + train photos	1,381 / 716	722 / 413	253 / 393
Littering / dumping	130 / 144	29 / 32	50 / 50
Loitering	282 / 41	85 / 13	64 / 11
Public sleep	49 / 25	14 / 7	12 / 6
Unattended items	416 / 74	137 / 181	720 / 242

A separate 806-photo smoking holdout is reported separately. Endpoints and adjacent frames remain correlated.

More data was acquired and screened for compatibility

Source diversity, duplicate control and label meaning matter more than download size

Source	Acquired / retained	Role and limit
Smoking photos + HMDB	1,118 photos; 906 clips	Photo learning; action-source transfer
Thermal loitering	874,936 observations; 31 dates	Provided tracks; one thermal camera
MEVA + public sleep	13 videos; 28 new-author photos	Action proxy; provisional photo labels
AVSS2007	3 videos; 2 clean primary events	One conservative source block
17 Oct author release	3 new clips + 3 ABODA aliases	30 highway-debris component frames

The latest release adds small-object stress cases, but adds no verified owner-associated unattended events.

Keep related sources together and freeze decisions

A source holdout is more informative than a random split of adjacent frames

Before evaluation

Preserve source / camera / author groups
Respect official video/photo partitions
Remove exact-pixel aliases
Purge known shared-film part aliases
Fit routes and thresholds on dev only

Limits that remain

Previously inspected tests are retrospective
Movie / actor identities are incomplete
Only a few physical cameras / events
Provisional interval / sleep labels
Repeated seeds add no new domains

No test-optimal threshold or lucky seed is substituted after seeing results.

Accuracy runs used available GPU compute

No pacing, CPU emulation or new latency benchmarks in the later studies

Quality execution

RTX 5000 Ada GPU
Batched CLIP extraction / head fitting
Deterministic VLM generation
Failures and abstentions stay counted
Pinned weights, inputs and raw scores

Numerical consistency

Fine-tuning: BF16 autocast, F32 weights
Final CLIP scoring: F32, TF32 off
Batch-shape sensitivity was audited
New detector: 830 F32 GPU views
Event delay measures task detection

Event onset delay is a quality metric; it is separate from computation latency.

Ranking, frame decisions and alerts are different

All three are needed to judge practical detection

Image / clip discrimination

AP and AUROC: score ranking
Recall + FPR: a fixed operating point
F1 + confusion counts: class balance
VLM hard labels: no invented AP
Weak candidates: prediction rate only

Continuous event detection

Past-only 20-second histories
Actual alert emission times
One matched alarm per labelled event
Normal false, repeated, unknown alerts
Onset delay + reviewed normal hours

An alert in an unreviewed transition is unknown; an early continuing alarm is not silently rematched to a later onset.

Expanded unattended images did not yield a clear winner

962 test endpoints • 242 positive / 720 normal • two camera families

Model	Precision	Recall	F1	FPR
MobileCLIP2-S2	25.3%	75.2%	0.379	74.6%
SigLIP2-B/16-224	0.0%	0.0%	0.000	2.6%
SmolVLM-500M	18.6%	63.6%	0.288	93.8%
MiniCPM-V 4.0	77.1%	34.7%	0.479	3.5%

MiniCPM has the best raw frame F1 here, while missing 65% of positive endpoints. Labels remain provisional.

Both CLIP visual backbones were updated

Two encoders × seeds 42 / 43 / 44 • frozen early layers and text towers

What learned

MobileCLIP: 18.1M visual parameters
SigLIP: 14.2M visual parameters
Last vision stage + five binary heads
Masked target losses and balancing
Scene-preserving augmentation

How overfitting was constrained

Pretrained feature / weight penalties
Frozen early layers and batch norms
Probe and pretrained-score controls
Development-selected routes / epochs
Three-seed variation retained

These six visual fine-tunes are distinct from later frozen-feature classifiers, GRUs and ensembles.

Fine-tuning helped some tasks and hurt others

Same-study pretrained score → development-selected primary route (AP)

Evidence	Cases	MobileCLIP AP	SigLIP AP
Smoking CCTV camera c1	646	.642 → .564	.669 → .650
Waste dumping clips	100	.424 → .858	.499 → .885
Loitering windows	75 (11 +)	.320 → .093	.097 → .097
Public-sleep photos	18 (6 +)	.502 → .976	.733 → .915
Unattended endpoints	962	.209 → .652	.309 → .896

Photo smoking improves strongly (.901 / .954 AP), but photo success did not transfer to CCTV.

The next screen combined data, crops and adaptation

33 configurations per target / backbone • five objectives • two CLIP encoders

Inputs and representations

Train: 3,258 → 3,973 examples
Automatic contextual person/item crops
Full / region visual representations
Temporal pooling vs ordered views
Sparse detection geometry

Candidate routes

Linear and nonlinear classifiers
Pretrained / adapted weight blends
Alpha: 0, .25, .5 and 1
Dev-source AP route selection
Raw, dev-F1 and FPR-budget cutoffs

330 configurations include references and 320 classifier fits; they are not 330 separate backbone fine-tunes.

Historical AP rose unevenly after the strategy screen

Previous selected bundle → new selected strategy • same previously inspected cases

Objective	MobileCLIP AP	SigLIP AP	Evidence limitation
Smoking CCTV	.564 → .690	.650 → .722	One held-out camera
Waste dumping	.858 → .806	.885 → .893	Full-clip action classification
Loitering	.093 → .489	.097 → .550	One Street Scene camera
Public sleep	.976 → .931	.915 → .883	Small provisional photo set
Unattended	.652 → .980	.896 → .925	Provisional overlapping endpoints

MobileCLIP dumping and both public-sleep photo routes regress. Higher AP still does not solve alert thresholds.

The broader transfer tests exposed remaining failures

Previous bundle → selected strategy AP • before later HMDB adaptation

Transfer evidence	Cases	MobileCLIP AP	SigLIP AP
New smoking-photo holdout	224	.953 → .997	.995 → 1.000
HMDB movie/web action clips	906	.605 → .576	.685 → .664
New-author public-sleep photos	28	.925 → .952	.943 → .933
MEVA abandonment-action proxy	27	.569 → .545	.637 → .621

Photo-domain gains coexist with movie/web and MEVA regressions. These sources are not interchangeable.

Loitering and public sleep remain data-limited

Their useful component results do not yet validate robust event detection

Loitering component

17,878 eligible supplied trajectories
Chronological 19 / 6 / 6-date split
Forest test AP .577 vs duration .283
Development AP .863
One thermal camera; no RGB tracking

Public-sleep component

28 new-author photo cases
MobileCLIP AP .952; SigLIP .933
Provisional visible-sleep labels
No confirmed persistence / duration
Rest, lying and bedding need controls

The next model must see the evidence the target requires: complete tracks or persistent state, with matched negatives.

Small people and missing items were concrete bottlenecks

117 deliberately selected errors were provisionally reviewed across the five targets

Smoking spatial evidence

Median smoker height: 11.2 px at 224
71.8% of positive frames below 12 px
Old crop misses half every target: 40.2%
Multi-person / upper crops: 0.3%
Coverage is separate from recognition

Item and ownership evidence

CAVIAR item recall at IoU .5:
Faster R-CNN tracks: 0 / 136
Grounding DINO tracks: 66 / 136
Proximity can confuse passers-by
Jitter can reset stationarity

Error-board findings are triage, not an independently reviewed estimate of failure causes.

Separate appearance from tracked event state

Automatic proposals → frozen CLIP features → causal heads → explicit alerts

1. Detect

Full scene + up to four persons
Whole / upper-body crops
Two item / owner contexts

2. Represent

Frozen CLIP visual features
Causal geometry and timestamps
Linear / GRU / instance heads

3. Decide

Past-only scores / same-item state
Persistence and alert emission
Event / normal / unknown counts

No annotation boxes choose inference crops. Missing owner history never becomes an invented ownership relation.

Fifty heads tested region, sequence and person evidence

8,931 modeling examples after the known-film source guard

Smoking: 14 heads / encoder

Scene and region linear baselines
Region GRU, three seeds
Region + geometry GRU, three seeds
Person-supervised MIL, three seeds
Hard-negative variant, three seeds

Unattended: 11 heads / encoder

Scene and region linear baselines
Region GRU, three seeds
COCO-geometry GRU, three seeds
Grounding-geometry GRU, three seeds
Unchanged visual CLIP features

Route and epoch selection maximize worst development-source AP, then mean AP. All seeds and regressions remain visible.

CCTV gains did not repeat consistently across backbones

205 HMDB evaluation clips now follow 530 grouped training clips; CCTV remains static

Model / source	Previous bundle	Pilot scene	Selected pilot
MobileCLIP / CCTV 646	.690	.626	.725
SigLIP / CCTV 646	.722	.717	.688
MobileCLIP / HMDB 205	.697	.842	.908
SigLIP / HMDB 205	.762	.888	.778

SigLIP’s development-selected pilot route regresses on CCTV and trails its own scene baseline on HMDB.

Ensembling improves HMDB; low-FPR CCTV remains hard

38 dev-only policies • every dev source must meet the 5% empirical FPR constraint

Policy	CCTV AP	Recall / FPR	HMDB AP	Recall / FPR
MobileCLIP / previous	0.725	16.5% / 7.9%	0.908	86.7% / 2.9%
MobileCLIP / 17 Oct	0.725	10.9% / 5.5%	0.908	86.7% / 2.3%
SigLIP / previous	0.688	2.3% / 1.6%	0.778	66.7% / 1.7%
SigLIP / 17 Oct	0.720	1.8% / 1.6%	0.964	76.7% / 1.7%

SigLIP selects the all-head ensemble. MobileCLIP retains the same person head with a stricter cutoff.

Explicit item state was more useful than a global score

Two clean AVSS events • 2.76 normal minutes • first used 16 Oct, now retrospective

Route / policy	Events	Normal false / repeats	Median onset delay
MobileCLIP / global score	2 / 2	1 / 3	25.8 s
SigLIP / global score	2 / 2	6 / 3	12.1 s
MobileCLIP + known state gate	2 / 2	0 / 0	36.3 s
SigLIP + known state gate	2 / 2	0 / 1	34.8 s
Known source state rule alone	2 / 2	0 / 0	34.8 s

The rule requires the same stationary item for 30 s and its previously associated owner absent for ≥0.5 s.

A good AP or zero observed alarms is not a guarantee

Insufficient normal footage is a measurable limitation

Historical calibration diagnostic

Entire normal recordings held aside
Both arms exclude those recordings
SigLIP: 0/4 → 1/4 events
Normal false alerts: 4 → 1
Still 26.3 false alerts / normal hour

Uncertainty in the AVSS result

2 events and 165.74 normal seconds
0 observed state-rule false alerts
95% one-sided Poisson upper bound:
65.1 false alerts / normal hour
Camera transfer remains untested

Even under a homogeneous Poisson assumption, about 3 zero-alert normal hours are needed for an upper bound below 1/hour.

More state machinery did not improve the AVSS result

Original detections and 30-second source rule are held fixed

Ablation	Events	Normal false / repeats	Median onset delay
Original global state debounce	2/2	0 / 0	34.82 s
Original tracks + item memory	2/2	0 / 0	34.82 s
Median stationarity + global	2/2	0 / 0	35.07 s
Median stationarity + memory	2/2	0 / 0	35.07 s

Keep the simpler state baseline: smoothing adds 0.25 s; item memory increases repeats on the censored HARD diagnostic.

The new highway clips are component stress tests

Pinned author release: three bundled clips, three explicit ABODA aliases

Acquisition and preparation

28.6 MB author archive
Three bundled highway videos
7.6–14.6 seconds each
Ten annotated positive frames / video
Actual FPS and timestamps checked

What cannot be inferred

Debris boxes do not establish an owner
No 30-second stationary evidence
No complete reviewed normal intervals
ABODA links add no independent data
No new unattended-event accuracy

The additional frames test detection of very small objects; they are not newly verified unattended events.

Fixed overlapping tiles test the small-object bottleneck

166 source-annotated frames × five views = 830 new GPU image views

Reference source	IoU	Full image recall	Full + four tiles recall
CAVIAR item source boxes	0.3	69/136 (50.7%)	117/136 (86.0%)
CAVIAR item source boxes	0.5	66/136 (48.5%)	116/136 (85.3%)
Highway-debris stress cases	0.3	0/30 (0.0%)	7/30 (23.3%)
Highway-debris stress cases	0.5	0/30 (0.0%)	7/30 (23.3%)

Same weights, query and score thresholds. Partial reference boxes support recall, not object precision or event accuracy.

VadCLIP remains a qualified smoking baseline

Offline supervised adaptation • different source groups and supervision

Measured test evidence

1,056 test frames
560 positive / 496 normal
AP 0.633; AUROC 0.587
Recall 51.8%; FPR 35.7%
F1 0.565, dev-frozen cutoff

Comparison boundaries

9 training clips / 4 validation clips
Adapter receives smoking supervision
Not an unsupervised five-task model
No new VadCLIP fine-tune this round
Not the same split as CLIP camera c1

Avoid a single leaderboard that mixes this offline supervised protocol with image-model zero-shot or event-stream tests.

No single model is ready to cover all five objectives

Use task-specific candidate routes and measure the evidence they actually consume

Objective	Useful candidate evidence	Main unresolved issue
Smoking	Person crops; SigLIP ensemble on HMDB	CCTV low-FPR recall; tiny smokers
Littering	SigLIP IWDD action AP .893	Pedestrian litter vs placing; localization
Loitering	Geometry / thermal supplied tracks	Automatic RGB tracks; unseen cameras
Public sleep	Photo probes / apparent posture	Verified persistent video; rest controls
Unattended	Grounded items + explicit state rule	Owner truth; detection gaps; normal hours

These are research candidates with different protocols; the AP values do not form a comparable deployment leaderboard.

Retain demonstrated gains and preserve regressions

Model selection must reward usable decisions as well as ranking

Useful progress

Real dumping-action supervision
Source-grouped data / duplicate checks
Person-region coverage for smoking
Explicit item persistence / owner state
Ensembles on adapted action clips

Not established or regressed

Photo gains → surveillance transfer
AP gains → stable alert thresholds
Extra GRU / mining → universal gains
More memory → fewer real alarms
More adjacent frames → more events

The latest event-policy ablations remain research alternatives; the simpler baseline is retained.

Prioritize independently labelled target-camera evidence

Freeze the next source holdout before looking at its predictions

Priority	Concrete work	Acceptance evidence
1. Data / labels	Review ownership, event boundaries, controls	Independent cameras / events, not frame count
2. Components	Validate crops, item recall and associations	No oracle crops; identity / coverage audits
3. Training	Train on confirmed source-balanced events	Fixed dev operating point; preserve controls
4. Final evaluation	New untouched cameras + long normal streams	Event recall, delay, duplicates and FAR bounds

Preserve an abstain / unknown state when duration, ownership or visibility is insufficient.

Every stage remains inspectable and replayable

Editable PowerPoint and vector PDF share one checked layout

Preserved research evidence

Fine-tuning: 46,926 held-out scores
Strategy screen: 117,612 scores
Video pilot: 99,528 scores
Latest: 5,643 smoking scores
Raw detector views + causal alert traces

Validation and deliverables

182 unit tests passed
Ruff lint / format + whitespace checks
38 policy selections checked
8 detector-metric cells replayed
PDF / PPTX + notes + source hashes

Historical artifacts remain unchanged. New results, qualifications and code are linked from the latest report.